

KUTSAL AKSU

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Education

University of Texas at Dallas

Aug 2026

Bachelor of Science in Software Engineering

Richardson, TX

- Relevant Coursework: Software Architecture, Data Structures & Algorithms, Software Testing, Database Systems
- Honors: Academic Excellence Scholarship

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, C

Frameworks & Databases: FastAPI, Node.js, PostgreSQL, pgvector, Pydantic, MongoDB, LangChain

Cloud & DevOps: AWS (EC2, Fargate, CDK, Bedrock, DynamoDB, S3), CI/CD (GitHub Actions), Linux, Git, Docker

AI & Testing: Claude Code, OpenAI Codex, RAG, LLM Evaluation, MCPs, Context Engineering, Pytest

Experience

AMC Health

May 2026 – Present

AI Software/Cloud Engineer

New York, New York

- Reduced test phone update latency from ~20 minutes to <5 seconds and prevented configuration data loss by replacing deployment-based configuration with a runtime-editable admin console backed by DynamoDB
- Eliminated silent AI voice-call failures by detecting livelocks within 5 turns, recovering calls to speech in ~3 seconds, and routing failed sessions into automated callbacks
- Improved patient address verification accuracy by redesigning spoken street-number parsing to correctly interpret mixed spoken numbers and prevent address mismatches in voice workflows

Secure IV AI

Jan 2026 – May 2026

Software Engineer, Team Lead | Senior Capstone

Richardson, TX

- Led 8-person engineering team building AuditOps, a compliance automation platform reducing SOC 2 audit preparation time from 200+ hours to <40 hours via automated evidence collection
- Developed agentic-AI system using RAG and prompt-engineering to analyze contracts and automatically determine compliance scope
- Designed document approval workflow with role-based access control, SHA-256 verification, and append-only audit logging
- Deployed compliance platform to AWS EC2 using FastAPI, React, and PostgreSQL with pgvector for vector similarity search

Sixfab

Sep 2025 – May 2026

Software Engineer

Allen, TX

- Built automated defect detection pipeline using Python and OpenCV, identifying anomalies in visual inspection data at 720 inspections/hour
- Deployed containerized application with Docker on Raspberry Pi, reducing setup time from 2 hours to 10 minutes across remote devices
- Built environment-based configuration system separating device parameters from code, enabling deployment across 10+ edge device profiles without modification
- Delivered technical demonstrations for the Alpon X5 AI Edge Computer at CES 2026; product received CES 2026 Best of Innovation Award (Enterprise Tech)

Projects

AWS Twilio Voice Assistant | *Python, FastAPI, AWS Bedrock (Nova Sonic), Twilio, DynamoDB, Docker, ECS Fargate*

- Built real-time AI voice agent connecting Twilio phone calls to Amazon Nova Sonic, supporting bidirectional audio streaming with one dedicated AI session per call
- Implemented observability pipeline with structured logging and 5 CloudWatch metrics for latency, errors, call volume, interruptions, and audio processing
- Containerized and deployed application to AWS ECS Fargate with automated infrastructure provisioning and load-balanced HTTPS endpoints